

Sec: XII_STU_IC_IIT_B1

Time: 3 Hrs.

PTM – 01

Date: 20-03-2023

Max. Marks: 300

JEE MAIN 2022 Model

PHYSICS

Section	Question Type	+Ve Marks	–Ve Marks	No. of Qs.	Total marks
Sec – I (Q.N : 1 – 20)	Questions with Single Correct Choice	4	–1	20	80
Sec–II (Q.N : 21 – 30)	Questions with Numerical Answer Type (+ / –Number)	4	–1	10	20
Total				30	100

CHEMISTRY

Section	Question Type	+Ve Marks	–Ve Marks	No. of Qs.	Total marks
Sec – I (Q.N : 31 – 50)	Questions with Single Correct Choice	4	–1	20	80
Sec–II (Q.N : 51 – 60)	Questions with Numerical Answer Type (+ / – Number)	4	–1	10	20
Total				30	100

MATHEMATICS

Section	Question Type	+Ve Marks	–Ve Marks	No. of Qs.	Total marks
Sec–I (Q.N : 61 – 80)	Questions with Single Correct Choice	4	–1	20	80
Sec–II (Q.N : 81 – 90)	Questions with Numerical Answer Type (+ / – Number)	4	–1	10	20
Total				30	100

Exam Syllabus

PHYSICS	Unit and Dimensions, motion in a straight line, Motion in plane, laws of motion, Work Power Energy, Collisions, System of particles and Rotational Motion and Gravitation
CHEMISTRY	Some basic concepts of Chemistry, Structure of atom, Classification of elements and Periodic properties, Chemical bonding and Molecular Structure, States of matter, Thermodynamics, Chemical and Ionic Equilibrium
MATHEMATICS	Sets, Basic Maths, Quadratic Equations, Trigonometric Ratios & Identities, Sequence and Series, Trigonometric Equations

PHYSICS

Max. Marks: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

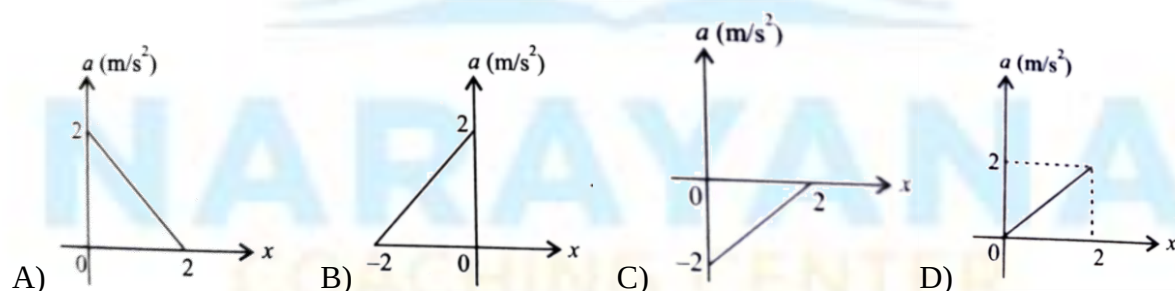
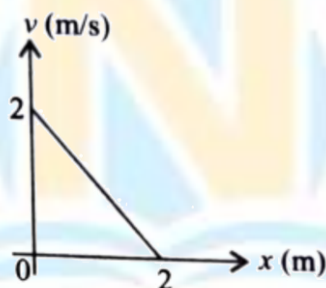
1. The potential energy of a particle is given by the expression $U(x) = -\alpha x + \beta \sin \frac{x}{\gamma}$. A dimensionless combination of the constants α, β and γ is

A) $\frac{\alpha}{\beta\gamma}$ B) $\frac{\alpha^2}{\beta\alpha}$ C) $\frac{\gamma}{\alpha\beta}$ D) $\frac{\alpha\gamma}{\beta}$

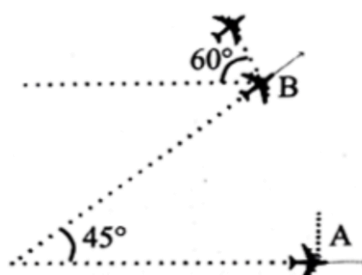
2. A car of mass m is travelling at a slow velocity v_0 . It is subjected to a drag resistance of the wind, which is proportional to its velocity, the constant of proportionality being k . The distance covered by the car before its speed becomes half is

A) $\frac{mv_0}{k}$ B) $\frac{2mv_0}{k}$ C) $\frac{mv_0}{2k}$ D) $\frac{mv_0}{4k}$

3. The velocity – displacement graph of a particle moving along a straight line is shown in figure. Then the acceleration – displacement graph is

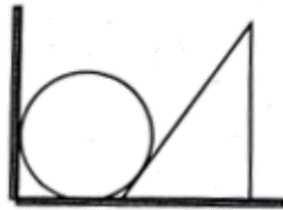


4. Passengers in the jet transport A flying east at a speed of 800 kmh^{-1} observe a second jet plane B that passes under the transport in horizontal flight. Although the nose of B is pointed in the 45° north east direction, plane B appears to the passengers in A to be moving away from the transport at the 60° angle as shown. The true velocity of B is

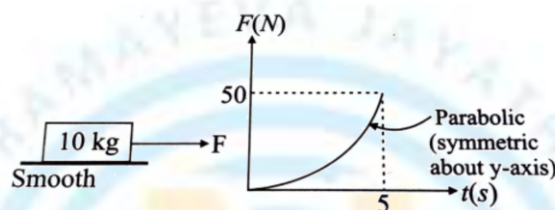


A) 586 kmh^{-1} B) $400\sqrt{2} \text{ kmh}^{-1}$ C) 717 kmh^{-1} D) 400 kmh^{-1}

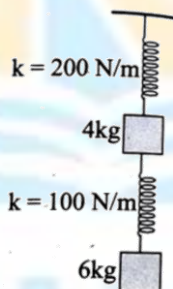
5. A sphere of radius 10 cm is in contact with a wedge as shown. The point of contact of the sphere is at height of 2 cm above the ground. If the wedge is moving leftwards with 20 cm/s, the sphere is moving with speed of



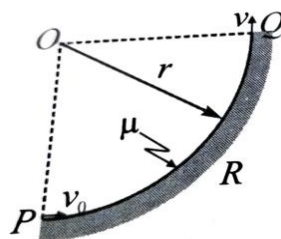
- A) 12 cm/s B) 15 cm/s C) 18 cm/s D) 21 cm/s
6. A force F is applied to the initially stationary cart. The variation of force with time is shown in the figure. The speed of cart at $t = 5$ s is



- A) 10 m/s B) 8.33 m/s C) 2 m/s D) zero
7. Two blocks of 4 kg and 6 kg are attached to springs, they are hanging in vertical position, lower spring breaks due to excessive force. Acceleration of 4 kg block just after breaking



- A) 15 m/s^2 B) 25 m/s^2 C) 10 m/s^2 D) Zero
8. In the figure shown PRQ is a curved vertical wall with rough inner surface while the floor is smooth. The radius of the curved wall is r and its length is quarter of a circle. A particle of mass m is projected on the floor at P, grazing the rough wall as shown, with an initial velocity v_0 . The velocity of the particle at Q will be $\left[\angle POQ = \frac{\pi}{2} \right]$

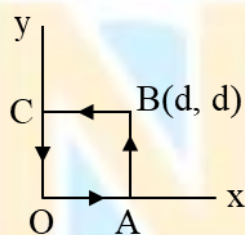


- A) $v = v_0 e^{-\mu\pi}$ B) $v = v_0 e^{\frac{-\mu\pi}{2}}$ C) $v = v_0 e^{\mu\pi}$ D) $v = v_0 e^{\frac{-\mu\pi}{r}}$

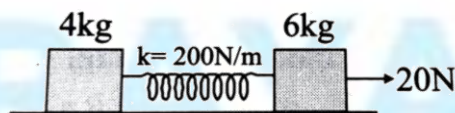
9. A particle of mass m oscillates along the horizontal diameter AB inside a smooth spherical shell of radius R . At any instant the kinetic energy of the particle is K . Then the force applied by the particle on the shell at this instant is



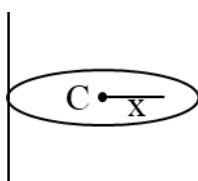
- A) $\frac{K}{R}$ B) $\frac{2K}{R}$ C) $\frac{3K}{R}$ D) $\frac{4K}{R}$
10. The work done by the force $\vec{F} = A(y^2\hat{i} + 2x^2\hat{j})$, where A is constant and x and y are in meters around the path shown is



- A) Zero B) Ad C) Ad^2 D) Ad^3
11. Two blocks of mass 4 kg and 6 kg are attached by a spring of spring constant $k=200\text{ N/m}$, both blocks are moving with same acceleration. Find elongation in spring.

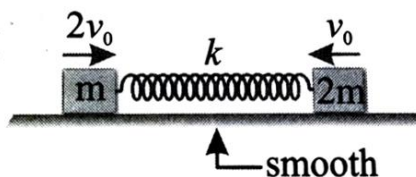


- A) 4 cm B) 10 cm C) 6 cm D) 2 cm
12. The density of a circular disc is given as $\sigma = \rho_0 x$ where ' x ' is the distance from the centre C . Its moment of inertia about an axis perpendicular to its plane and passing through its edge shown in figure is.

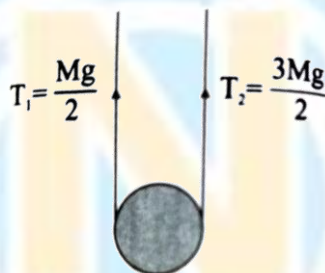


- A) $\frac{15}{16}\rho_0\pi R^5$ B) $\frac{16}{15}\rho_0\pi R^5$ C) $\frac{6}{5}\rho_0\pi R^5$ D) $\frac{5}{6}\rho_0\pi R^5$

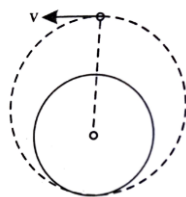
13. Two blocks A and B of masses m and $2m$ respectively, attached at opposite ends of a spring of force constant k , placed on smooth horizontal surface. Spring is initially at its natural length l . A is given a velocity $2v_0$ and B given velocity v_0 as shown. The maximum separation between $2m$ and center of mass of the system will be



- A) $\frac{\ell}{3} + \sqrt{\frac{8mv_0^2}{k}}$ B) $\frac{2\ell}{3} + \sqrt{\frac{2mv_0^2}{k}}$
C) $\frac{\ell}{3} + \sqrt{\frac{2mv_0^2}{3k}}$ D) None of these
14. A uniform disc of mass M and radius R is lifted using a string as shown in the figure. Then

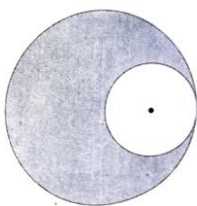


- A) Its linear acceleration is g upward
B) Its rate of change of angular momentum is MgR
C) Its angular acceleration is $\frac{2g}{R}$
D) All of these
15. A particle of mass m is projected in a transverse direction from a height R (Radius of earth) above surface of earth. Find minimum initial speed so that it just grazes the surface of earth [Mass of the earth M]

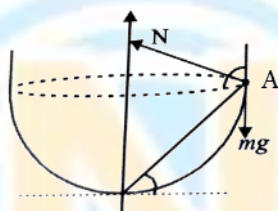


- A) $\sqrt{\frac{3GM}{2R}}$ B) $\sqrt{\frac{2GM}{3R}}$
C) $\sqrt{\frac{GM}{R}}$ D) $\sqrt{\frac{GM}{3R}}$

16. From a solid sphere of mass M and radius R , a spherical portion of radius $R/2$ is removed, as shown in the figure. Taking gravitational potential $V = 0$ at $r = \infty$, the potential at the centre of the cavity thus formed is (G = gravitational constant)

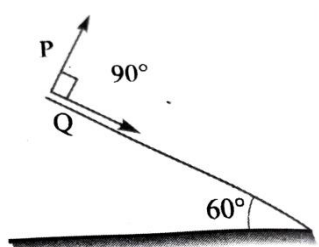


- A) $-\frac{GM}{R}$ B) $-\frac{2GM}{3R}$ C) $-\frac{2GM}{R}$ D) $-\frac{GM}{2R}$
17. In the given figure, a smooth parabolic wire track lies in the xy -plane (vertical). The shape of track is defined by the equation $y = x^2$. A ring of mass m which can slide freely on the wire track, is placed at the position $A(1,1)$. The track is rotated with constant angular speed ω such there is no relative slipping between the ring and the track. The value of ω is



- A) $\sqrt{g/2}$ B) $\sqrt{g}$ C) $\sqrt{2g}$ D) $2\sqrt{g}$
18. Three particle A, B and C start from the origin at the same time, A with velocity 'a' along x -axis, B with velocity 'b' along y -axis and C with velocity 'c' in xy -plane along a line $x = y$. The magnitude of c , so that they always remain collinear is

- A) $\frac{a+b}{2}$ B) $\sqrt{ab}$ C) $\frac{ab}{a+b}$ D) $\frac{\sqrt{2}ab}{a+b}$
19. Planck's constant h , speed of light c and gravitational constant G are used to form a unit of length l and a unit of mass M . Then the correct option (s) is (are)
- A) $M \propto \sqrt{c}$ B) $L \propto \sqrt{G}$
C) $L \propto \sqrt{h}$ D) All of these
20. A particle P is projected from a point on the surface of long smooth inclined plane and Q starts moving down the plane from the same position. P and Q collide after 4 second. The speed of projection of P is : ($g = 10 \text{ m/s}^2$)



- A) 5m/s B) 10 m/s C) 15m/s D) 20 m/s

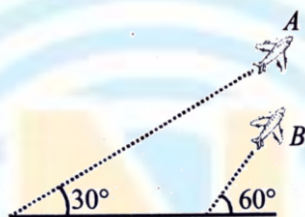
SECTION-II

(Numerical Value Answer Type)

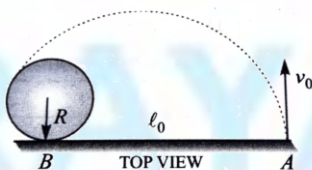
This section contains 10 questions. The answer to each question is a Numerical value. If the Answer is in the decimals, **Mark the nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

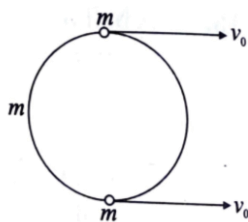
21. Airplanes A and B are flying with constant velocity in the same vertical plane at angles 30° and 60° with respect to the horizontal respectively as shown in figure. The speed of A is $100\sqrt{3}$ m/s. At time $t = 0$ s, an observer in A finds B at a distance of 500 m. The observer sees B moving with a constant velocity perpendicular to the line of motion of A. If at $t = t_0$ A just escapes being hit by B, t_0 in seconds is



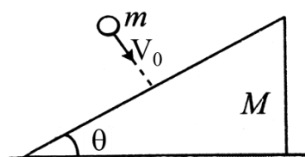
22. A horizontal plane supports a stationary vertical cylinder of radius R and a disc A attached to the cylinder by a horizontal thread AB of length ℓ_0 . An initial velocity v_0 is imparted to the disc as shown in fig. The time it move along the plane till it strikes the cylinder is the time $t = \frac{\ell_0^2}{xRv_0}$. Find the value of x . The friction is assumed to be absent.



23. Two small beads each of mass m are attached to a smooth ring of mass m as shown in the figure. The whole system is kept on smooth horizontal table. Initially, each bead has given velocity v_0 as shown in figure. If the magnitude of velocity of bead just before impact is xv_0 , the value of x is nearest value



24. A ball of mass m collides perpendicular on a smooth stationary wedge of mass M . The coefficient of restitution of collision is e . If the velocity of the wedge after collision is $V = \frac{m(1+e)V_0 \sin \theta}{M + m \sin^2 \theta}$. The value of x is



25. A shell of mass $(m_1 + m_2)$ is fired with a given velocity in a given direction. At the highest point of its path, the shell explodes into two fragments of mass m_1 and m_2 . The explosion produces an additional kinetic energy E and the fragments separate in a horizontal direction. If the horizontal distance between the masses on the ground at which they hit the ground, (if vertical component of initial velocity is v_0) is $\frac{v_0}{g} \sqrt{x E \left(\frac{1}{m_1} + \frac{1}{m_2} \right)}$. Find the value of x .
26. A uniform rod of mass $m = 5.0$ kg and length $L = 90$ cm rests on a smooth horizontal surface. One of the ends of the rod is struck with the impulse $J = 3.0$ N s in a horizontal direction perpendicular to the rod. As a result the rod obtains the momentum $p = 3.0$ N s. Find the force with which one half of the rod will act on the other in the process of motion. (in N)
27. A bob of mass m , suspended by a string of length ℓ_1 , is given a minimum velocity required to complete a full circle in the vertical plane. At the highest point, it collides elastically with another bob of mass m suspended by a string of length ℓ_2 , which is initially at rest. Both the strings are mass-less and inextensible. If the second bob, after collision acquires the minimum speed required to complete a full circle in the vertical plane, the ratio $\frac{\ell_1}{\ell_2}$ is
28. The gravitational field in a region is given by $\vec{E} = -(10\hat{i} + 10\hat{j})$ N/kg. Find the work done by an external agent to slowly shift a particle of mass 2 kg from the point (0,0) to a point (5 m, 4 m).
29. A planet of mass m moves along a circle around the sun with a velocity $v = 34.9$ km/s (relative to the heliocentric reference frame). Find the period of revolution around the sun. (in earth days)
30. Two masses m_1 and m_2 at an infinite distance from each other are initially at rest, start interacting gravitationally. If their velocity of approach when they are at a distance r apart is $\sqrt{\frac{x G (m_1 + m_2)}{r}}$. Find the value of x .

CHEMISTRY
Max. Marks: 100
SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

31. Two sucrose solutions 125 ml of 1.50 M $C_{12}H_{22}O_{11}$ and 275 ml of 1.25 M $C_{12}H_{22}O_{11}$ are mixed. Assuming a final solution of volume of 400 ml. What is the molarity of $C_{12}H_{22}O_{11}$ in the final solution ?
(A) 1.635 M (B) 1.468 M (C) 1.328 M (D) 1.375 M
32. 12 g of Mg (At. Mass = 24) will react with an acid (HCl) to give:
(A) 1.0 mole of $MgCl_2$ (B) 2.0 mole of H_2
(C) 1.5 mole of H_2 (D) 0.5 mole of $MgCl_2$
33. The oxidation number of sulphur in S_8 , S_2F_2 , H_2S and H_2SO_4 respectively are
(A) 0, +1, -2 and 6 (B) +2, 0, +2 and 6 (C) 0, +1, +2 and 4 (D) -2, 0, +2 and 6
34. Photoelectric effect is the phenomenon in which?
(A) Photons come out of a metal when it is hit by a beam of electrons
(B) Photons come out of the nucleus of an atom under the action of an electric field
(C) Electrons come out of a metal with a constant velocity which depends on the frequency and intensity of incident light wave
(D) Electrons come out of a metal with different velocities not greater than a certain value which depends only on the frequency of the incident light wave and not on its intensity
35. If λ_0 is the threshold wavelength for photoelectric emission, λ the wavelength of light falling on the surface of a metal and m is the mass of electron, then the velocity of ejected electron is given by
(A) $\left[\frac{2h}{m} (\lambda_0 - \lambda) \right]^{1/2}$ (B) $\left[\frac{2hc}{m} (\lambda_0 - \lambda) \right]^{1/2}$ (C) $\left[\frac{2hc}{m} \left(\frac{\lambda_0 - \lambda}{\lambda_0 \lambda} \right) \right]^{1/2}$ (D) $\left[\frac{2hc}{m} \left(\frac{1}{\lambda_0} - \frac{1}{\lambda} \right) \right]^{1/2}$
36. The first ionization energy of Ar is less than that of Ne. An explanation of this fact is that
(A) the effective nuclear charge experienced by a valence electron in Ar is much larger than that in Ne
(B) the effective nuclear charge experienced by a valence electron in Ar is much smaller than that in Ne
(C) the atomic radius of Ar is larger than that of Ne
(D) the atomic radius of Ar is smaller than that of Ne
37. The coordination geometry around Boron atom in BF_3 and hybridization of Boron
(A) tetrahedral, sp^3 (B) tetrahedral, sp^3 (C) planar, sp^2 (D) linear, sp^2

38. Which ones among CO_3^{2-} , H_2O , BF_3 and NO_3^- have planar structure?

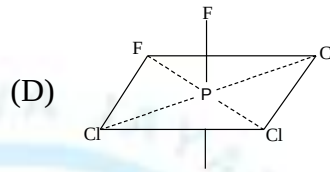
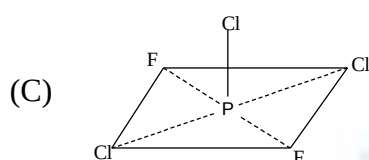
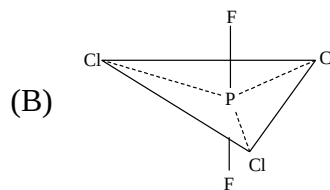
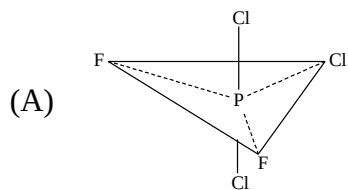
(A) CO_3^{2-} and BF_3

(B) H_2O and NO_3^-

(C) H_2O and CO_3^{2-}

(D) H_2O and BF_3

39. The correct structure of PCl_3F_2 is



40. Match List I (species) with List II (Hybridisation) and select the correct answer using the codes given

Below

List – I

A. BCl_3

B. NH_3

C. PCl_5

D. XeF_4

List – II

a. sp^3

b. $\text{sp}^3 \text{d}^2$

c. sp^2

d. $\text{sp}^3 \text{d}$

Codes

	A	B	C	D
(A)	c	a	d	b
(B)	a	c	d	b
(C)	a	b	d	c
(D)	c	a	b	d

41. A gas X diffuses three times faster than another gas Y, the ratio of their $\left(\frac{d_X}{d_Y}\right)$ densities is (under similar condition of T & P)

(A) $\frac{1}{9}$

(B) $\frac{9}{1}$

(C) $\frac{1}{3}$

(D) $\frac{\sqrt{3}}{1}$

42. Two flask A and B of 1L capacity each contains SO_2 and H_2 gases respectively maintained at 300 K and 2 atm. If number of H_2 moles in flask B is N, the total number of atoms in flask A will be

(A) $2 N_A \times N$

(B) $N_A \times N$

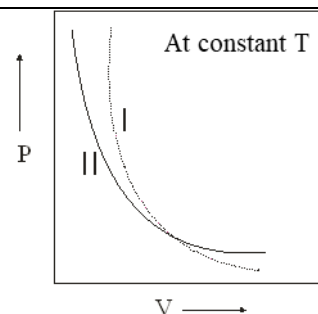
(C) $1.5 N_A \times N$

(D) $3 N_A \times N$

43. Consider the curve I and II in the following figure

Select the correct statement

- (A) Curve I and II represents ideal and real gases respectively
(B) Curve I and II represents real and ideal gases respectively
(C) Curve I and II represents real and real gases respectively
(D) Curve I and II represents ideal and ideal gases respectively

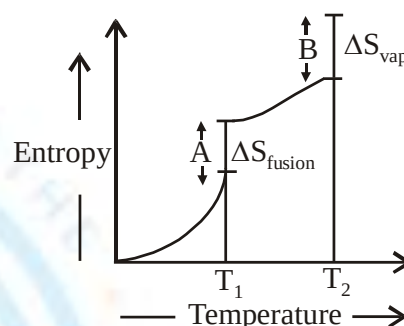


44. The molar heats of combustion of $C_2H_2(g)$, C (graphite) and $H_2(g)$ are -310.62 , -94.05 , -68.32 kcal respectively. The heat of formation of acetylene is

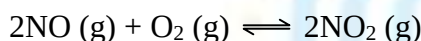
- (A) $+54.2$ kcal (B) -54.2 kcal (C) $+68.3$ kcal (D) -68.3

45. Observe the following curve for a substance and identify the correct statement

- (A) T_1 is melting point, T_2 is boiling point
(B) T_2 is melting point, T_1 is boiling point
(C) T_1 is melting point, T_2 is inversion temperature
(D) T_1 freezing point and T_2 is boiling point



46. What is the equilibrium constant (K_c) for the following system?



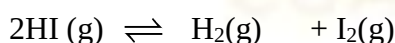
At a given temperature the equilibrium concentrations of the gases are $[NO] = 0.890$ M, $[O_2] = 0.250$ M and $[NO_2] = 0.0320$ M.

- (A) 5.17×10^{-3} (B) 4.6×10^{-2} (C) 7.62×10^{-3} (D) 1.68×10^{-4}

47. In the dissociation of HI into H_2 and I_2 , 20% of HI is dissociated at equilibrium. The K_p for the reaction $HI \rightleftharpoons \frac{1}{2}H_2 + \frac{1}{2}I_2$ is

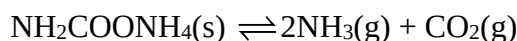
- (A) 0.125 (B) 0.0156 (C) 0.0286 (D) 0.0125

48. Which of the following is correct relation of degree of dissociation (α) and K_p for the following reaction?



- (A) $\alpha = \frac{2\sqrt{K_p}}{(1+\sqrt{K_p})}$ (B) $\alpha = \frac{2\sqrt{K_p}}{(\sqrt{1+2K_p})}$ (C) $\alpha = \frac{\sqrt{K_p}}{(1+2\sqrt{K_p})}$ (D) $\alpha = \frac{2\sqrt{K_p}}{(1+2\sqrt{K_p})}$

49. Solid ammonium carbamate dissociated according to the following equation



At equilibrium ammonia gas added in the system till the partial pressure of ammonia becomes equal to initial equilibrium pressure. The ratio of total pressure now (new equilibrium) to initial equilibrium pressure is

- (A) $\frac{31}{9}$ (B) $\frac{31}{27}$ (C) $\frac{1}{9}$ (D) $\frac{57}{23}$

50. The pH of a weak monobasic acid is given as

(A) $p^H = \frac{1}{2}(pK_a + \log C)$

(B) $p^H = \frac{1}{2}(pK_b - \log C)$

(C) $p^H = 2(pK_a - \log C)$

(D) $p^H = \frac{1}{2}(pK_a - \log C)$

SECTION-II

(Numerical Value Answer Type)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer is in the decimals, **Mark the nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

51. When $K_2Cr_2O_7$ is converted to K_2CrO_4 , the change in the oxidation state of chromium is

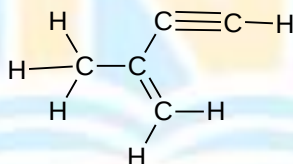
52. A photon of 300 nm is absorbed by a gas and then remits two photons. One remitted photon has wavelength 496 nm, the wavelength of second remitted photon is

53. In an atom, the total number of electrons having quantum numbers

$$n = 4, |m_1| = 1 \text{ and } m_s = -\frac{1}{2} \text{ is}$$

54. How many atoms in CO_2 are linearly arranged?

55. Consider the following molecule



How many carbon atoms are sp -hybridized?

56. Dipole moment of HCl molecule is found to be 0.816 D. Assuming HCl bond length is equal to be $1\text{\AA}$, the % ionic character of HCl molecule is (in %)

57. 50 mL of a gas A diffuse through a membrane in the same time as for the diffusion of 40 mL of a gas B under identical pressure and temperature conditions. If molar mass of A is 64, that of B would be

58. Boiling point of a liquid is 50 K at 1 atm and $\Delta H_{\text{vap}} = 460.6 \text{ cal mol}^{-1}$. What will be its boiling point at 10 atm? (in K)

59. Equilibrium constant for a reaction $A \rightleftharpoons B$ is 4. If concentration of reactants increases by three times, the new equilibrium constant becomes

60. The pH of 0.005 M H_2SO_4 solution is ($\log 5 = 0.699$)

MATHEMATICS
Max. Marks: 100
SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

61. The maximum sum of the series $\left(20 + \frac{1}{4}\right) + \left(18 + \frac{2}{4}\right) + \left(16 + \frac{3}{4}\right) + \left(14 + \frac{4}{4}\right) + \left(12 + \frac{5}{4}\right) + \dots$ is
- A) $\frac{255}{2}$ B) $\frac{495}{4}$ C) $\frac{253}{2}$ D) $\frac{223}{2}$
62. The solution set of the equation $\left|\frac{x}{x-1}\right| + |x| = \frac{x^2}{|x-1|}$ is
- A) $[0, \infty)$ B) $(-\infty, \infty)$ C) $(1, \infty) \cup \{0\}$ D) $(-\infty, 0]$
63. If the remainders of the polynomial $f(x)$ when divided by $x+1, x+2, x-2$ are 6, 15, 3 then the remainder of $f(x)$ when divided by $(x+1)(x+2)(x-2)$ is
- A) $2x^2 - 3x + 1$ B) $3x^2 - 2x + 1$ C) $2x^2 - x - 3$ D) $3x^2 - 2x + 1$
64. If α, β, γ are the roots of $x^3 + qx + r = 0$, then equation whose roots are $\frac{\beta+\gamma}{\alpha^2}, \frac{\gamma+\alpha}{\beta^2}, \frac{\alpha+\beta}{\gamma^2}$ is
- A) $rx^3 + qx^2 - 1 = 0$ B) $rx^3 - qx^2 - 1 = 0$ C) $qx^3 + rx^2 + 1 = 0$ D) $qx^3 - rx^2 - 1 = 0$
65. If $\tan^2 \theta = \sqrt{3} + (\sqrt{3} - 1)\tan \theta$, where $\theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ then $\theta =$
- A) $\left\{-\frac{\pi}{3}, \frac{\pi}{4}\right\}$ B) $\left\{\frac{\pi}{3}, -\frac{\pi}{4}\right\}$ C) $\left\{\frac{\pi}{6}, -\frac{\pi}{3}\right\}$ D) $\left\{-\frac{\pi}{12}, \frac{\pi}{12}\right\}$
66. The solution of the equation $\cos^{103} x - \sin^{103} x = 1$
- A) $-\frac{\pi}{2}$ B) $\frac{\pi}{4}$ C) $\frac{\pi}{2}$ D) π
67. The general value of θ satisfying $\sin \theta = \frac{1}{2}, \tan \theta = \frac{1}{\sqrt{3}}$ is
- A) $2n\pi + \frac{\pi}{6}; n \in \mathbb{Z}$ B) $2n\pi - \frac{7\pi}{6}; n \in \mathbb{Z}$ C) $2n\pi + \frac{5\pi}{6}; n \in \mathbb{Z}$ D) $n\pi + \frac{\pi}{6}; n \in \mathbb{Z}$
68. The value of $\frac{3 + \cot 76^\circ \cdot \cot 16^\circ}{\cot 76^\circ + \cot 16^\circ}$ is
- A) $\tan 46^\circ$ B) $\cot 46^\circ$ C) $\tan 92^\circ$ D) $\sqrt{3}$
69. If α, β, γ are solutions in $\left(0, \frac{\pi}{2}\right)$ for the equation $\sin 7\theta + \sin 4\theta + \sin \theta = 0$ then $\cos(\alpha + \gamma) =$ where, $(\alpha < \beta < \gamma)$
- A) $\frac{1}{2}$ B) $-\frac{1}{2}$ C) 0 D) 1

70. The sum of first hundred numbers common to the two A.P's 12, 15, 18, and 17, 21, 25, is
 A) 56100 B) 65100 C) 61500 D) 72300
71. If $x = \sum_{n=0}^{\infty} a^n, y = \sum_{n=0}^{\infty} b^n, z = \sum_{n=0}^{\infty} c^n$ where a, b, c are in AP and $|a| < 1, |b| < 1, |c| < 1$, then x, y, z are in
 (a,b,c are distinct)
 A) AP B) GP C) HP D) AGP
72. The set $(A \cup B \cup C) \cap (A \cap B^c \cap C^c)^c \cap C^c$ is equals to
 A) $B \cap C$ B) $A \cap C$ C) $A \cap B$ D) $B \cap C^c$
73. If $a_1, a_2, a_3, \dots, a_n$ are positive real numbers whose product is a fixed number C , then the minimum value of $a_1 + a_2 + a_3 + \dots + a_{n-1} + 2a_n$ is
 A) $n(2C)^{1/n}$ B) $(n+1)C^{1/n}$ C) $2nC^{1/n}$ D) $(n+1)(2C)^{1/n}$
74. Let a, b, c are the A.M.s between two positive numbers p and q such that $a + b + c = 15$ and α, β, γ are H.M.s. between the same two numbers such that $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{5}{3}$, then (p, q) can be
 A) (2, 8) B) (9, 2) C) (9, 1) D) (3, 7)
75. Sum of n terms of the series $ab + (a+1)(b+1) + (a+2)(b+2) + \dots + (a+(n-1))(b+(n-1))$ if $ab = \frac{1}{6}$ and $(a+b) = \frac{1}{3}$, is
 A) $\frac{n}{6}(1-2n)^2$ B) $\frac{n}{6}(1+n-2n^2)$ C) $\frac{n}{6}(1-2n+2n^2)$ D) $\frac{n(n^2-1)}{6}$
76. The solution set of the system of equations $x + y = \frac{2\pi}{3}; \cos x + \cos y = \frac{3}{2}$ is ; $x, y \in \mathbb{R}$
 A) $x = -n\pi + \frac{2\pi}{3}; y = n\pi; n \in \mathbb{Z}$ B) ϕ
 C) $x = n\pi; y = -n\pi + \frac{2\pi}{3}; n \in \mathbb{Z}$ D) $x = 2n\pi; n \in \mathbb{Z}$
77. The equation $e^{\sin x} - e^{-\sin x} - 4 = 0$ has
 A) Exactly one real root B) Exactly four real roots
 C) Infinite number of real roots D) no real roots
78. If the roots of the equation $x^2 - 2ax + a^2 + a - 3 = 0$ are real and less than 3, then which of the following is correct
 A) $a < 2$ B) $2 \leq a \leq 3$ C) $3 < a \leq 4$ D) $a > 4$
79. If $a, a_1, a_2, a_3, \dots, a_{2n}, b$ are in A.P and $a, g_1, g_2, g_3, \dots, g_{2n}, b$ are in G.P and 'h' is H.M of a and b then $\frac{a_1 + a_{2n}}{g_1 g_{2n}} + \frac{a_2 + a_{2n-1}}{g_2 g_{2n-1}} + \dots + \frac{a_n + a_{n+1}}{g_n g_{n+1}}$ is equal to
 A) $\frac{2n}{h}$ B) $2nh$ C) nh D) $\frac{n}{h}$

80. Sum to n terms of the sequence : $\frac{1}{1^3} + \frac{1+2}{1^3+2^3} + \frac{1+2+3}{1^3+2^3+3^3} + \dots$ is

(A) $\frac{n}{n+1}$

(B) $\frac{2n}{n+1}$

(C) $\frac{2(n+2)}{n+1}$

(D) $\frac{2n^2}{n+1}$

SECTION-II

(Numerical Value Answer Type)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer is in the decimals, **Mark the nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

 81. If roots of the equation $x^2 - 10cx - 11d = 0$ are a, b and those of $x^2 - 10ax - 11b = 0$ are c, d then the value of $\left(\frac{a+c}{11} - 11\right)$ must be equal to (a, b, c , and d are distinct numbers)

 82. If $\cos \theta + \cos^2 \theta + \cos^3 \theta = 1$ and $\sin^6 \theta = a + b \sin^2 \theta + c \sin^4 \theta$ then $a + b + c =$

 83. If $\frac{\tan x}{2} = \frac{\tan y}{3} = \frac{\tan z}{5}$ and $x + y + z = \pi$, $\tan^2 x + \tan^2 y + \tan^2 z = \frac{38}{K}$ then $|K| =$

 84. If $\left(\frac{1}{\sec^2 \alpha - \cos^2 \alpha} + \frac{1}{\cos^2 \alpha - \sin^2 \alpha}\right) \sin^2 \alpha \cos^2 \alpha = \frac{1}{3}$ then the value of $2 \cos^2 \alpha =$

 85. If α, β are the roots of $4x^2 - 15x + \lambda = 0$ where $\lambda \in \mathbb{R}$, such that $1 < \alpha < 2$ and $2 < \beta < 3$. Then the number of integral values of λ is

 86. Let α, β are roots of $x^2 + 12x - 9 = 0$. If $a_n = \alpha^n + \beta^n$ for $n \geq 1$, then the value of $\sum_{n=1}^{10} \left(\frac{9a_n - a_{n+2}}{2a_{n+1}}\right) =$

 87. If $\sin^2 x - 2 \sin x - 1 = 0$ has exactly four different solutions in $x \in [0, n\pi]$ then minimum value of 'n' can be ($n \in \mathbb{N}$)

 88. Suppose $A_1, A_2, \dots, A_{30}$ are 30 sets each having five elements and $B_1, B_2, \dots, B_n$ are n sets each with 3 elements such that $\bigcup_{i=1}^{30} A_i = \bigcup_{j=1}^n B_j = S$. If each element of S belongs to exactly ten of the A_i 's and exactly 9 of the B_j 's then the value of n is

 89. If α, β, γ are such that $\alpha + \beta + \gamma = 2$, $\alpha^2 + \beta^2 + \gamma^2 = 6$, $\alpha^3 + \beta^3 + \gamma^3 = 8$, then $\alpha^4 + \beta^4 + \gamma^4$ is

 90. If 3 positive real numbers a, b, c are in A.P with $abc = 4$ then the least possible value of $[b]$ is (where $[.]$ represents greatest integer function)

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